



ZACHERY TAYLOR

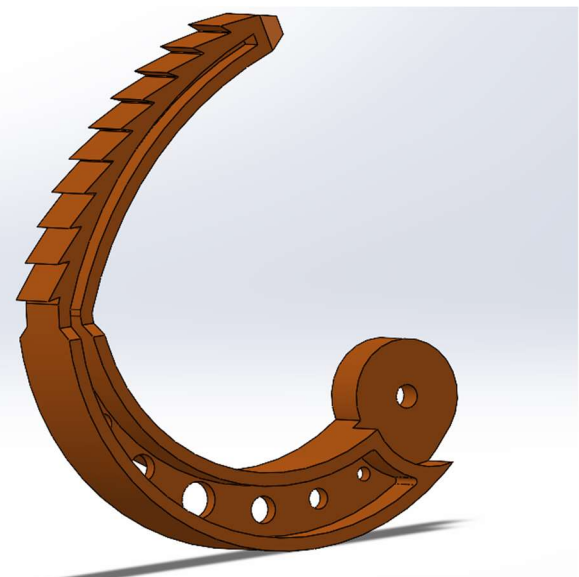
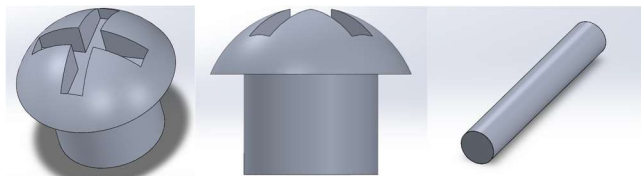
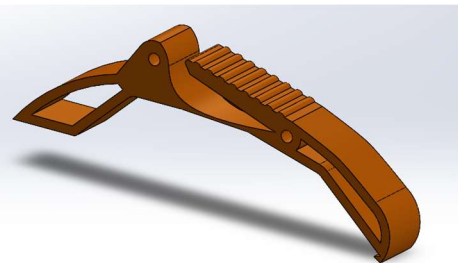
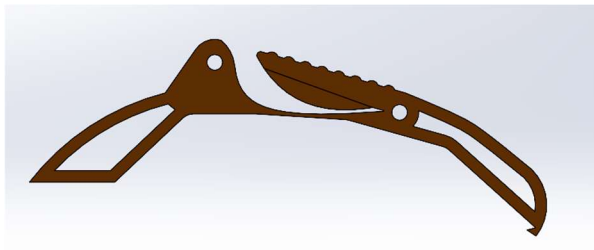
EML 3022

SUMMER 2023

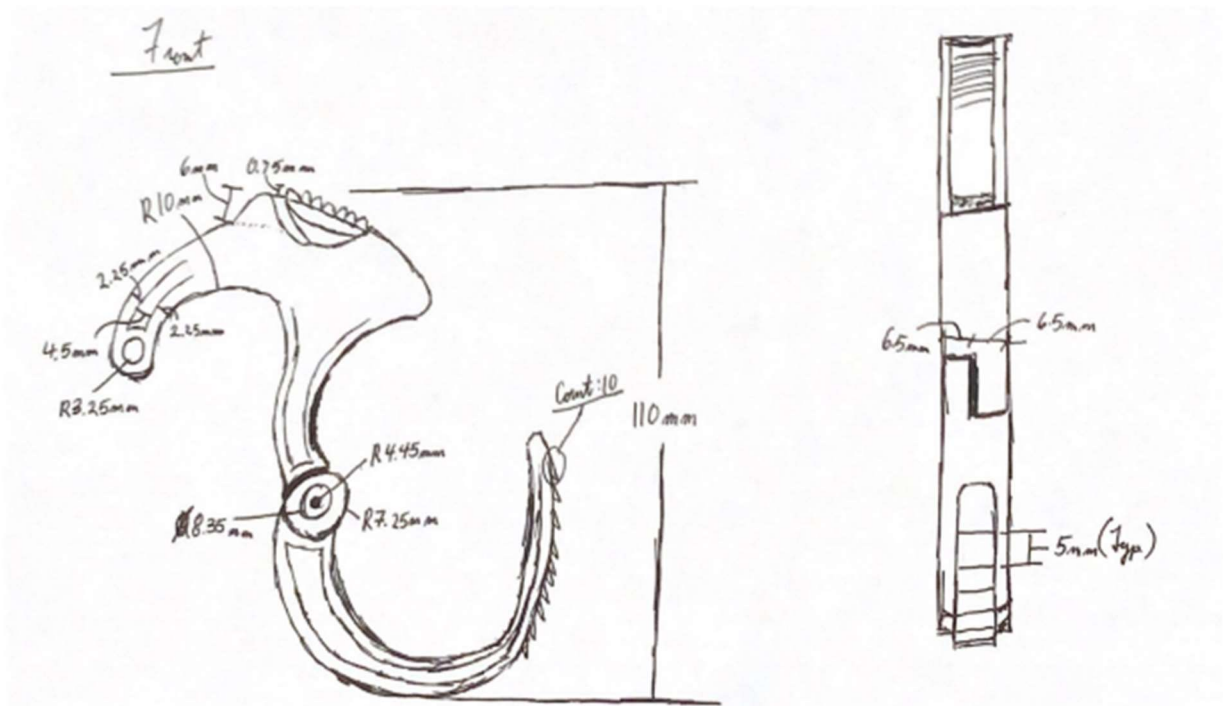
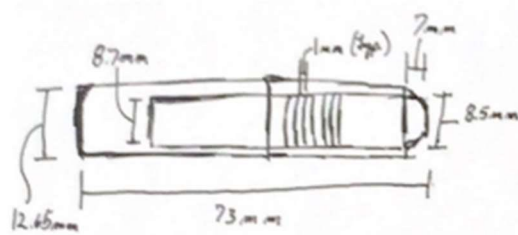
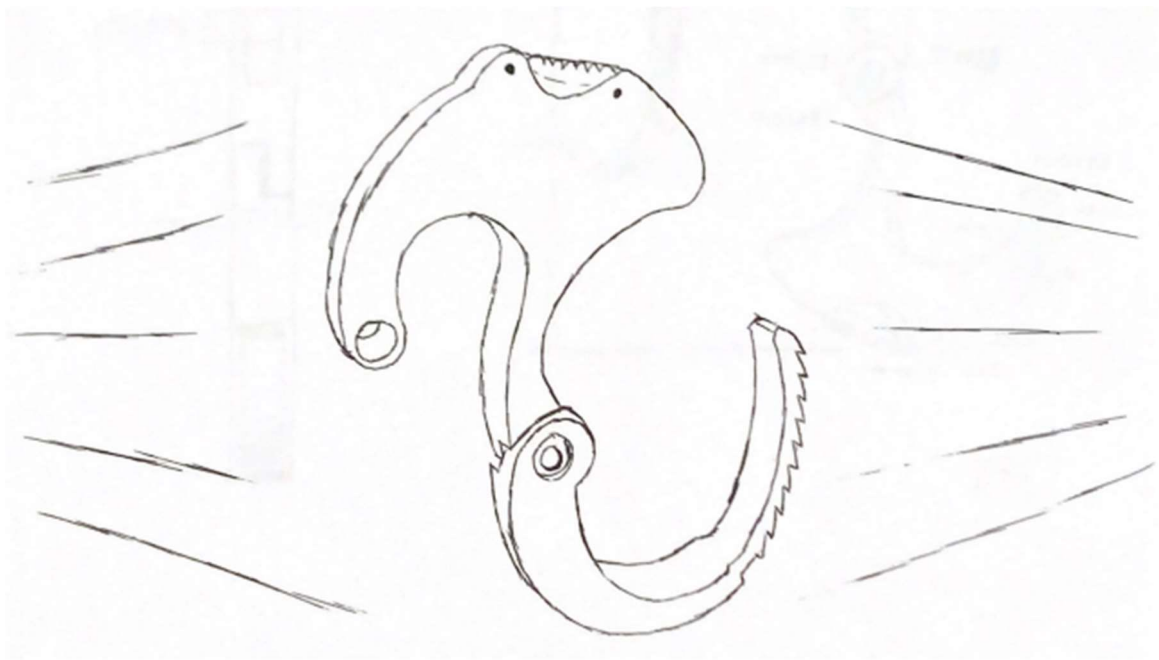
AUGUST 4TH



This is the CableCuff PRO, a reusable cable management tool primarily used to keep cords, cables and ropes organized. I use this handy device on a weekly basis, due to the variable sizes offered, to keep everything from my phone cord to my garden hose from getting folds, kinks, or harsh bends in the line. Per the design description, it is made of polycarbonate/ABS polymers in the black and orange pieces as well as stainless steel pins, washers, screws. Compared to other competitive cable management tools, this one provides the best balance of cost, ease of use, and reusability that is unrivaled. The tool also conveniently includes a hanging loop as well as what I like to refer to a 'holding hook'. This holding hook is ergonomic and allows for ease of closing, opening, and holding of the tool even with loaded up with cords. The ribbed extrusion in the top opening of the base is a standalone part that allows, through flexing, to allow the teeth to be released. I have modified the base piece (black) to have a shorter solid base to reduce weight and in-turn most likely reducing cost. I have also adjusted the holding hook to allow for stronger pulling strength than before, as well as an even more ergonomic feel in the hand.

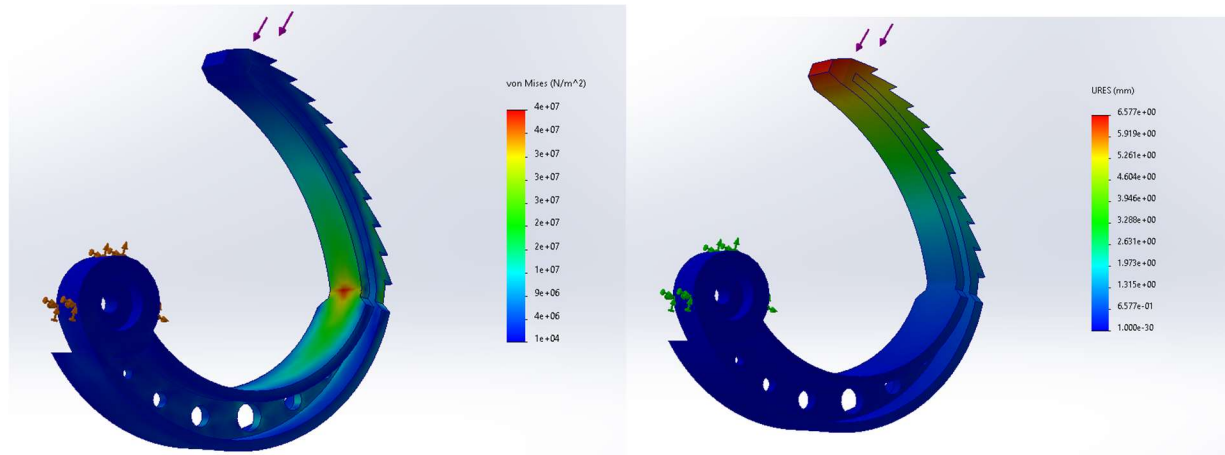




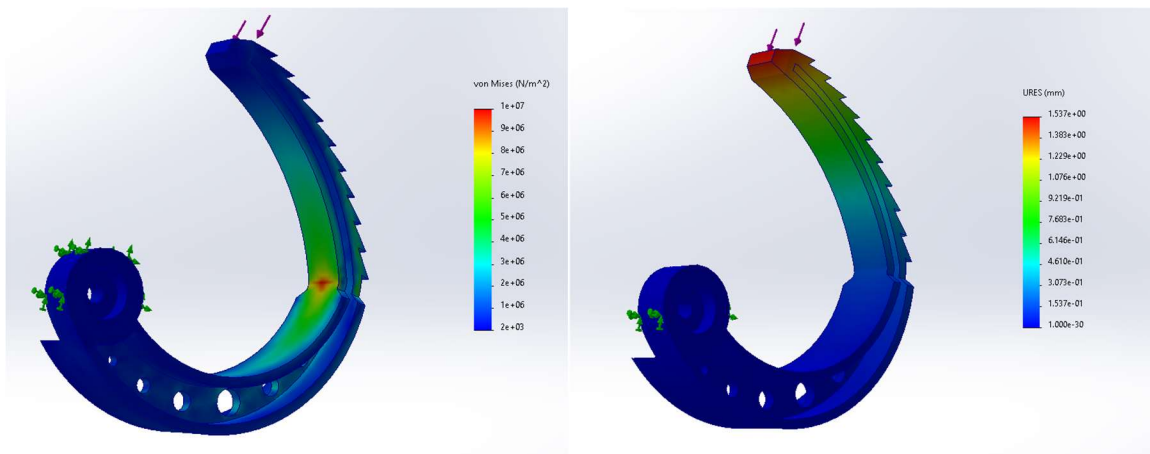


ANALYSIS

OLD

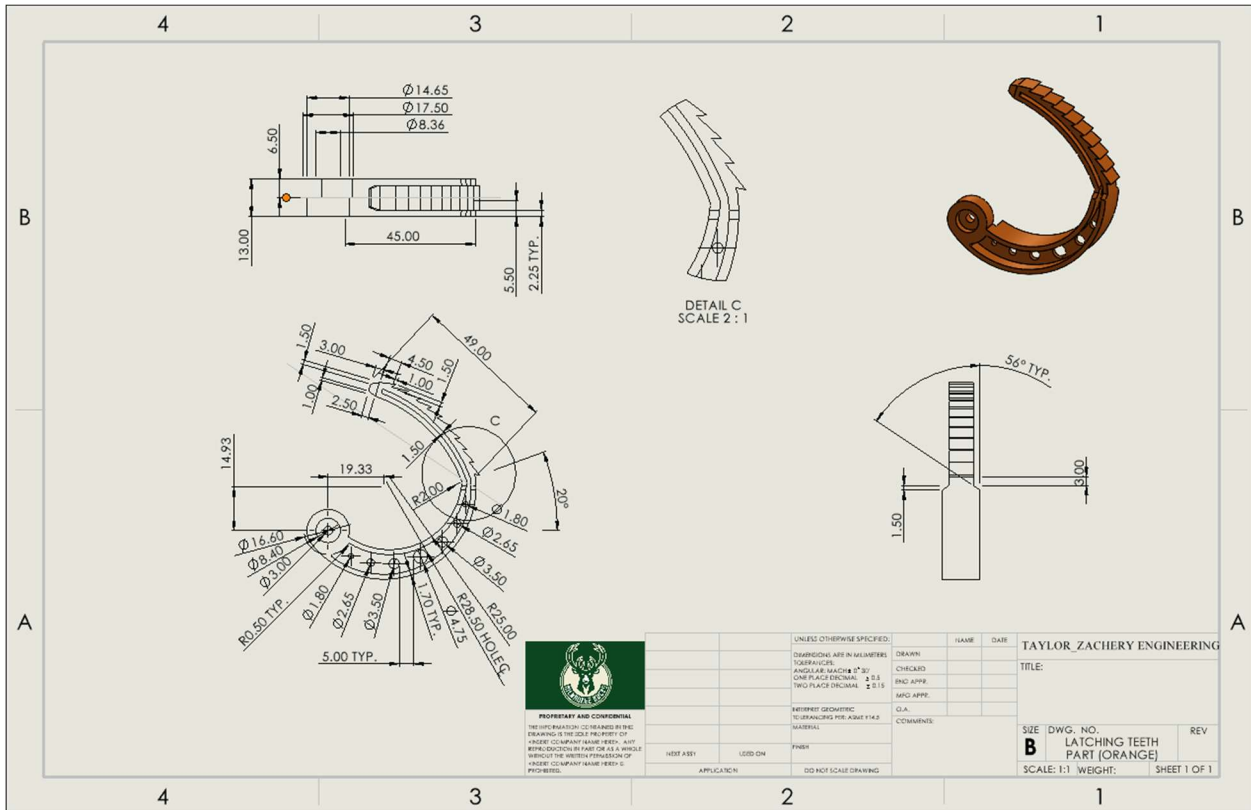


NEW



Visually it is hard to see the improvements to design but choosing to use Epoxy, Unfilled for the new and improved version rather than polycarbonate/ABS polymers I see a slight increase in Factor of Safety as well as an overall stronger part. I chose to put force on the top of the arc, at the end of the teeth line due to these seeming to be the weakest point I've encountered when using the CableCuff PRO over time. The point of red stress lines up exactly where I've had breaking/snapping issues in the past with the tool.

DRAWING



MASS PROPERTIES



	(in ³)	(lb)
Measured	1.30	.05
Theoretical	1.28	.05
% Error	1.54%	0.00%
Standard Deviation	0.0141	0.00

Based on the very high-tech ability of water displacement I calculated that the existing CableCuff PRO has a volume roughly 1.3 in³. I got this number from a cup conversion to cubic inches, with an observed measurement of 0.09 Cups. For the weight my scale would not pickup the weight of the tool without something else on the scale, therefore I created a constant. The yellow Dickie's cup filled with H₂O (water) weighs exactly 1.60 lbs, once adding the CableCuff PRO the scale reads exactly 1.65 lbs. My theoretical tool came out to 1.28 in³ and 0.5 lbs. I know didn't use the EXACT correct material but for weight purposes it looks like I was pretty close!

